QUANTUM
SOURCE

Quick Brand Guide

Introduction

Company Description

Quantum Source is developing a novel quantum computing architecture for fault-tolerant, commercially viable, useful quantum computers that can scale to millions of qubits at a fraction of the complexity, footprint, power consumption, and cost of other technologies and implementations. Quantum Source's architecture uniquely combines atoms and photons to construct complex 3D cluster states, serving as the backbone of error- correction codes. By allowing for the deterministic generation of small clusters of entangled photons, Quantum Source addresses the scalability and cost efficiency challenges , which is essential for making quantum computing feasible.

The Brand

The quantum computer industry is evergrowing with numerous players elbowing their way to the headlines. This is a no frills industry, cutting edge, where directness is key. The brand's challenge is to project QS"s unique offering in a confident, sophisticated and compelling way, presenting it as a viable frontrunner and most importantly making sure not to box the brand in, leaving it room to grow and evolve further down the line.

Key Words

Visionaries
Deterministic
Magnitude
Realistic
Achievable
Quality
Revolutionary
Authentic

Resonator
Entanglement
Photon-atom
Efficient
Light
Accuracy
Fault tolerant
Quantum

Speed
Innovation
Cutting edge
Viable
Fronttrunners
Strategic
Delay loop
Powerful

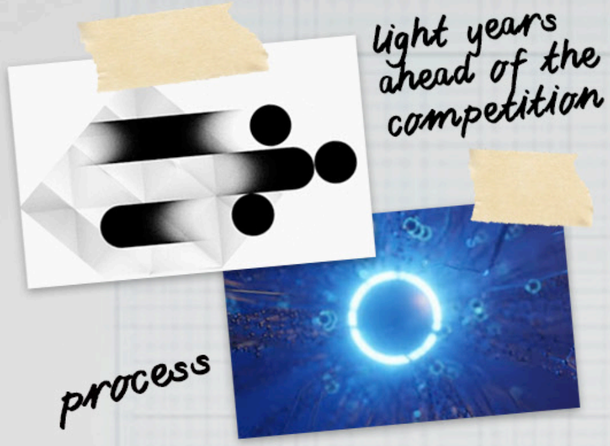
Affordable
Clusters
Quantity
Qubits
Experience
Pioneering
Descrete
Scalable

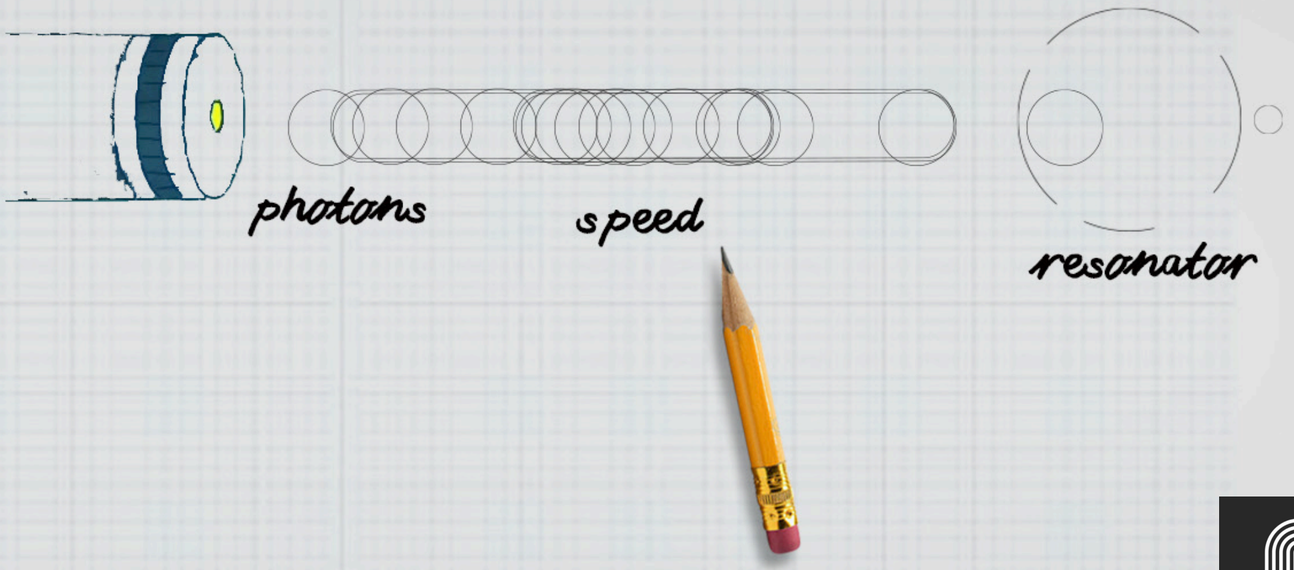
Wave guide
Tapestry
Stitcher
Optics

The narrative

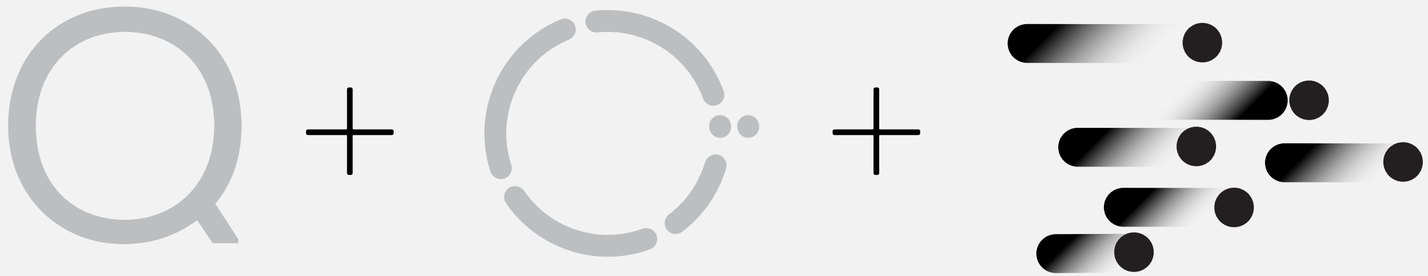
Exploring the visual landscape, drawing inspiration from the elements central to the process, simplifying them visually and in tandem embuing a sense of unrivalled expertise that projects confidence and certainty that Quantum Source's technological knowhow is unquestionably ahead of the game.



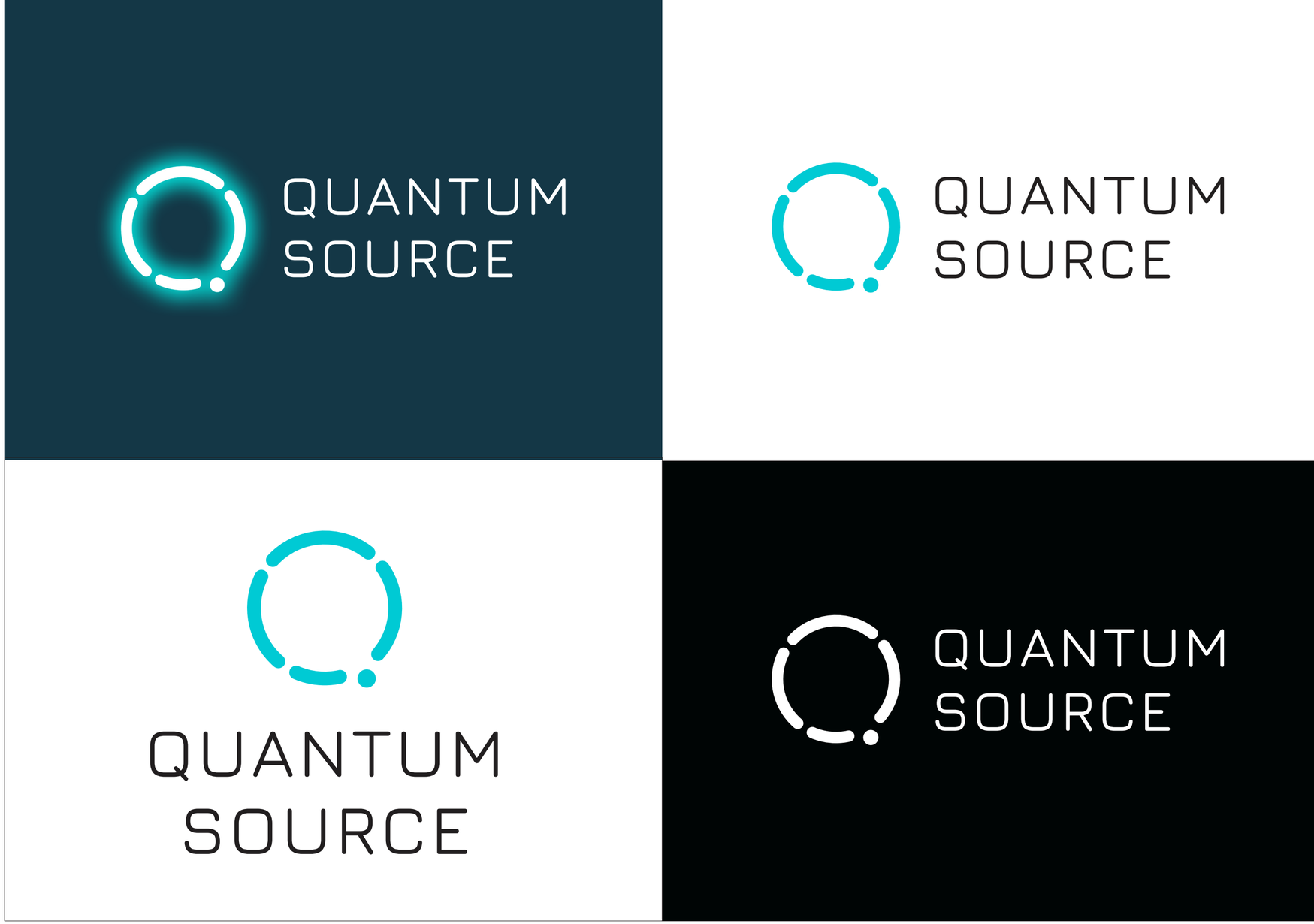




The Narrative



Logo


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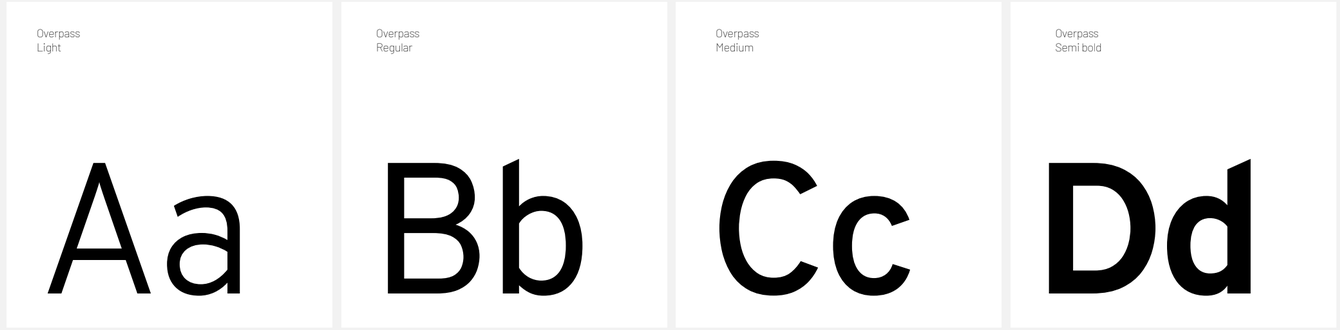
Typography

Logo font - Jura

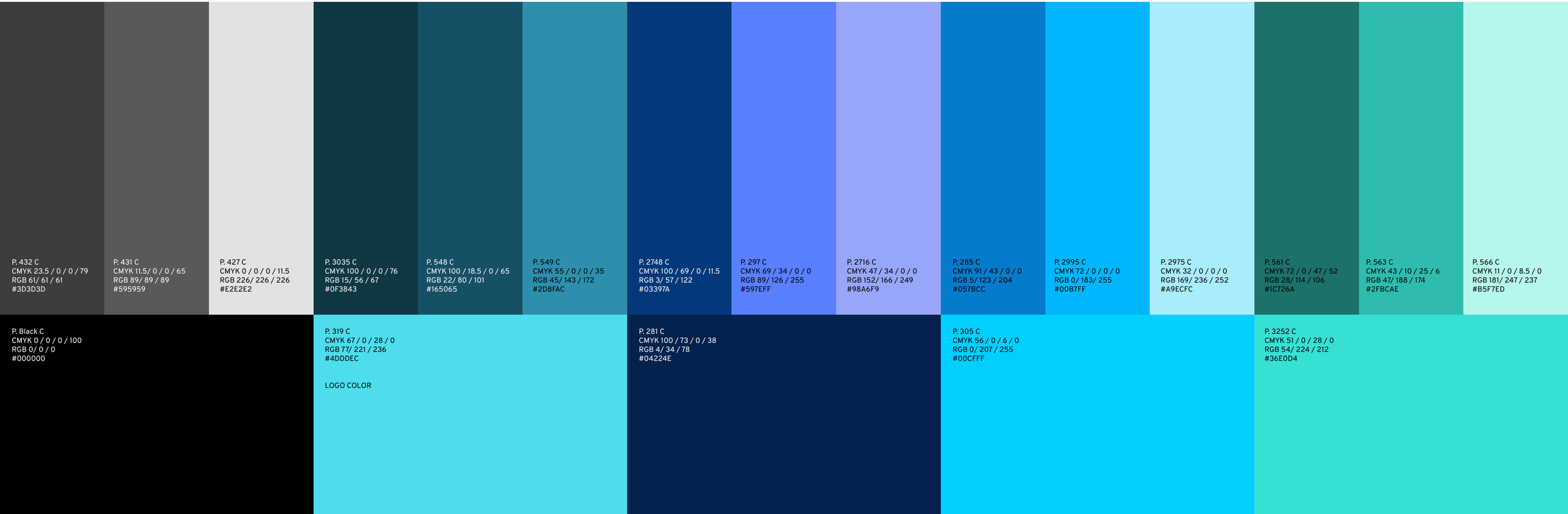
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Body text - Overpass

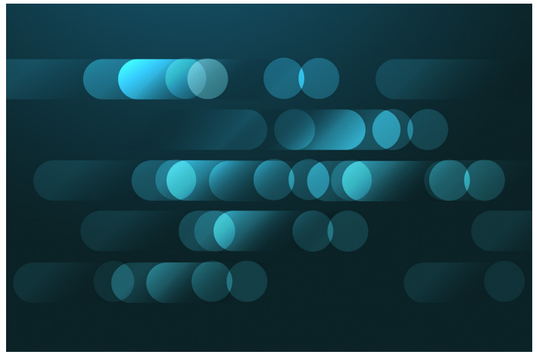
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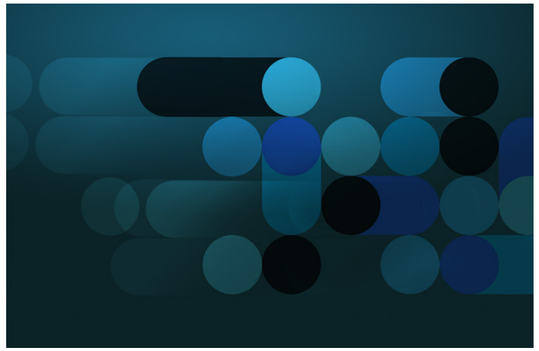
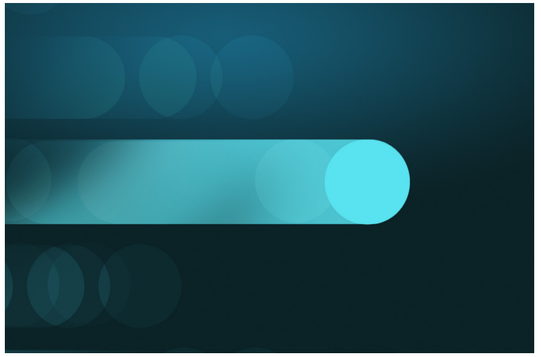
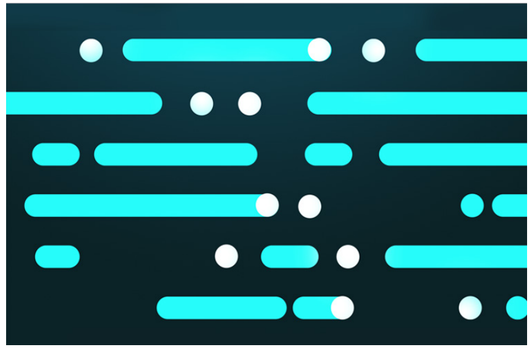


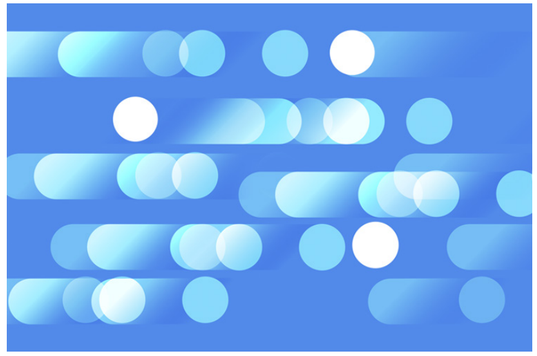
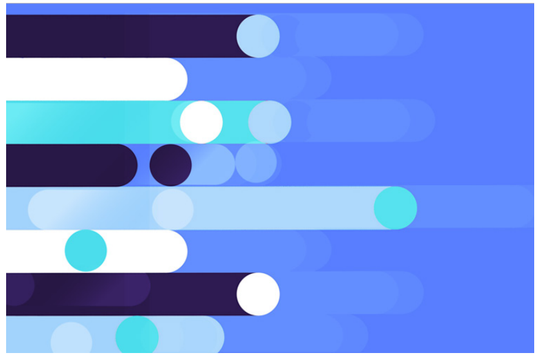
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CMYK 11.5/ 0 / 0 / 65
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CMYK 0 / 0 / 0 / 11.5
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RGB 54/ 224 / 212
#36E0D4

Patterns



This might be knocked out




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37
Rb
Rubidium
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BRINGING
QUANTUM
COMPUTING
TO LIGHT
QUANTUM SOURCE
QUANTUM
SOURCE
BRINGING QUANTUM COMPUTING TO LIGHT


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Revolutionizing photonic quantum computing.

Making scalable useful quantum computers a reality

Founding Team

ODED MELAMED
CEO

GIL SEMO
VP R&D

PROF. BARAK DAYAN
CHIEF SCIENTIST

DAN CHARASH
CHAIRMAN

World-Class Team

Management
Experienced leadership team that built and led successful companies, from the idea incubation stage to successful exits.

altair · Provigent · anobit

Top Scientists and Engineers
Top-notch team in quantum optics and silicon photonics technologies from some of the leading academic institutes worldwide.

MIT · Caltech · Yale

Industry Experience
Years of experience working for the top companies in their domain.

SONY · broadcom · elop

QS by numbers

Quantum Source was established in 2021

$27M seed round from 5 VC's making it one of the largest seed financings in QC

39 employees

18 hold PhD in Physics or EE

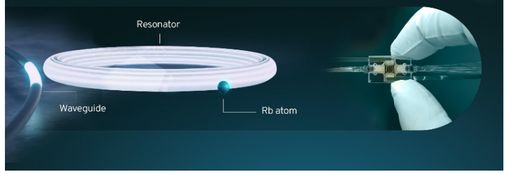
6 post-doctoral in some of the leading academic institutes

4 professors

backed by: pitango · ECLIPSE · GROVE VENTURES · 10D · DELL TECHNOLOGY CAPITAL

Solving the challenge of RSG gen

- The use of micron-scale optical cavities enables coupling single photons with single atom
- Enable deterministic generation of single-photons and entangling gate
- Manufactured in standard fab



Unlocking the scale-up bottlenecks

Improvement of
>45,000x

	Parameter	PsiQuantum SFWM	Cavity QED	Quantum Source Improvement
Photon Source Purity: 99.8% Loss <10%	Success probability	2%	95%	x100
	Units needed	100	1	
		Linear optics	Cavity QED	
GHZ (3 entangled qubits) Fault error: < 0.5% Loss: < 10%	Success probability	3%	95%	x450
	GHZ units needed	75	1	
	Sources units needed	75	1	
		Linear optics	Cavity QED	
RSG	GHZ units needed	36	36	x1
		Linear optics	Cavity QED	
Computer	RSG units needed	200	200	x1
Total Sources		324M	7,200	

Large-scale, fault tolerant QC are needed to solve meaningful problems

Breaking RSA
30 millions qubits
(6,000 perfect qubits)

Molecule Simulation
5 millions qubits
(1,000 perfect qubits)

Derivative Pricing
37 millions qubits
(7,500 perfect qubits)

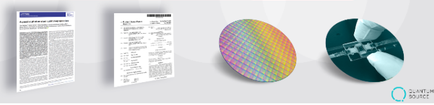
Where we are today

Concept demonstrated
Prof. Dayan's Lab (2018)

Strong IP position
US 10,304,535 granted
- 6 patent applications submitted
- 15 patents applications in the works (priority date May 2022)

Implementation using commercial SiN process
Three tapeouts done, two versions already delivered and operational May 2022)

System ready for silicon evaluation
- MOT is operational and atom delivery system is ready
- On-chip atom trapping on chip



Roadmap

2023
Atom Trapped
Atom trapped in pre- determined location near

2024
Photonic Entanglement
Photons entanglement using cavity-QED technology

2025
Resource State Generator (RSG)
Resource state for large-scale QC >1,000,000 qubits

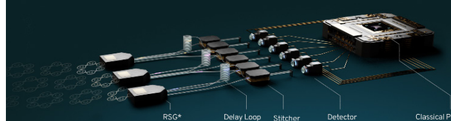
2028
1 million qubit QC

Deterministic

- Orders of magnitude more efficient compared to existing technologies
- No complex cooling is required
- Can scale up to hundreds of millions of qubits



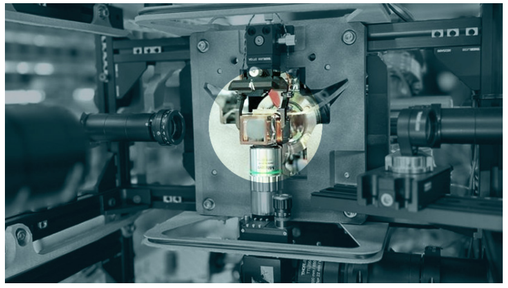
Modular architectures for fault tolerant photonics quantum computing



*RSG = Resource state generator
Resource State Generator
Generating small cluster of entangled photons (~10)

Probabilistic





Existing solutions will not be able to solve meaningful problems

1st Generation
Small noisy < 100 qubits

Google · TeenQ · rigetti · ColdQuanta · AQT · Amazon · OSC · Honeywell · IQM · Alpine · Oxford

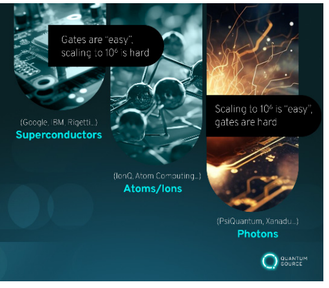
1.5 Generation
Small noisy < 1000 qubits

IBM · PASQAL · qci · QuEra · oxford ionics

2nd Generation
Fault-tolerant <millions of qubits

XANADU · Quantum Circuits, Inc. · atom computing · PsiQuantum · Quantum Machines

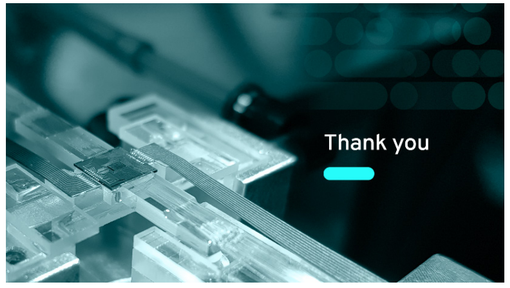
Leading candidate technologies for quantum computing



(Google, IBM, Rigetti...)
Superconductors

(IonQ, Atom Computing...)
Atoms/Ions

(PsiQuantum, Xanadu...)
Photons






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